

SoundCloud Clone [MODULES & FEATURES]

Module 1: Authentication & User Management

- **Registration & Verification:** Email-based registration with CAPTCHA and automated verification/resend workflows.
- **Account Recovery:** Self-service password reset and email update triggers.
- **Social Identity:** One-click Google/social login integration.
- **OAuth Flow:** Secure authorization using SoundCloud's current standard.
- **JWT & Refresh Tokens:** Industry-standard secure token handling for persistent sessions.

Module 2: User Profile & Social Identity

- **Profile Customization:** Dynamic bio, location, and "Favorite Genres" tagging.
- **Account Tiers:** Logic to distinguish between **Artist** (uploader) and **Listener** roles.
- **Visual Assets:** Management of avatars and high-resolution cover photos.
- **Web Profiles:** Ability to link external social profiles (e.g., Instagram, Twitter, personal website).
- **Privacy Control:** Public vs. private profile visibility settings.

Module 3: Followers & Social Graph

- **Relationship Management:** Real-time follow/unfollow system with automatic feed updates.
- **Network Lists:** Dedicated views for Followers, Following, and "Suggested Users."
- **Moderation:** User blocking and unblocking logic with a managed "Blocked Users" list.

Module 4: Audio Upload & Track Management

- **Multi-Format Support:** Upload and storage for MP3, WAV, and high-bitrate audio.
- **Metadata Engine:** Title, genre, descriptive tags, and release date management.
- **Transcoding Logic:** Automatic handling of track state (Processing vs. Finished).
- **Track Visibility:** Toggle tracks between Public (searchable) and Private (link-only).
- **Waveform Generation:** Visual representation of audio peaks (Mock or generated).

Module 5: Playback & Streaming Engine

- **High-Fidelity Streaming:** Core player with Play, Pause, Seek, and Volume control.
- **Playback Accessibility:** Logic to handle **Playable**, **Preview**, or **Blocked** states (Region or Tier based).
- **User History:** "Recently Played" and "Listening History" tracking.
- **Responsive Player:** Sticky/persistent player UI that works seamlessly on Web and Mobile.

Module 6: Engagement & Social Interactions

- **Favorites & Likes:** One-tap liking of tracks with a global "Favoriters" count.
- **Reposts:** Social sharing of tracks to a user's own feed/profile.

- **Timestamped Comments:** Ability to leave comments at specific seconds in the audio waveform.
- **Engagement Lists:** View list of users who Liked or Reposted a specific track.

Module 7: Sets & Playlists

- **Playlist CRUD:** Create, edit, and delete collections of tracks (Sets).
- **Track Sequencing:** Drag-and-drop reordering and "Add/Remove" functionality.
- **Playlist Privacy:** Secret vs. Public playlists with unique shareable "Secret Tokens."
- **Embed Support:** Generate simple iframe codes for sharing playlists externally.

Module 8: Feed, Search & Discovery

- **Stream/Activity Feed:** Chronological feed of new tracks from followed artists.
- **Resource Resolver:** Feature to resolve standard permalinks (URLs) into internal resource IDs.
- **Global Search:** Advanced search across Tracks, Users, and Playlists using keyword matching.
- **Trending & Charts:** Discovery logic based on recent play counts and engagement velocity.

Module 9: Messaging & Track Sharing

- **1-to-1 Direct Messaging:** Private text communication between users.
- **In-Chat Previews:** Embeddable track/playlist cards within the message thread.
- **Status Tracking:** Unread message counts and message-specific blocking rules.

Module 10: Real-Time Notifications

- **Activity Triggers:** Instant alerts for new Followers, Likes, Reposts, and Comments.
- **State Management:** Global "Mark as Read" and unread notification counter.
- **Push Notifications:** Mobile-ready alerts for time-sensitive social actions.

Module 11: Moderation & Admin Dashboard

- **Report System:** User-facing flags for Copyright or Inappropriate Content.
- **Admin Panel:** Content management tools to hide/remove tracks and suspend accounts.
- **Platform Health:** Analytics dashboard (Total active users, play-through rates, and storage usage).

Module 12: Premium Subscription (Pro/Go+)

- **Paywall Logic:** Enforce upload limits (e.g., 3 tracks for Free users vs. Unlimited for Pro).
- **Stripe Integration:** Mocked payment processing for subscription lifecycles.
- **Premium Perks:** Ad-free experience and mock "Offline Listening" (downloading) capabilities.